

Playwright Capstone Project Design Document

Application Under Test: Practice Software Testing

Project Overview

Field	Details
Project Title	Playwright Automation Framework for Practice Software Testing
Website	https://practicesoftwaretesting.com
Framework	Playwright + JavaScript
Design Pattern	Page Object Model (POM)
Reporting	Allure Reports
CI/CD	GitHub Actions
Browsers	Chromium, Firefox, WebKit
Total Modules	10
Estimated Test Cases	140

Website Selection

The Practice Software Testing website was selected because it provides real-world e-commerce workflows, stable selectors, responsive design, API compatibility, and automation-friendly architecture suitable for enterprise-grade Playwright automation frameworks.

Project Objectives

- Develop an enterprise-level Playwright automation framework using JavaScript and the Page Object Model (POM).
- Automate around 140 test cases across 10 modules including authentication, search, cart, checkout, API, and responsive testing.
- Validate both UI and backend functionalities using Playwright UI automation and API testing features.
- Implement cross-browser, responsive, and parallel test execution for Chromium, Firefox, and WebKit browsers.
- Generate detailed Allure Reports with screenshots, traces, execution history, and failure analysis.
- Integrate the framework with GitHub Actions CI/CD and demonstrate industry-standard SDET automation practices.

Project Architecture

Playwright-Capstone/

```
|
├─ tests/
├─ pages/
├─ fixtures/
├─ utils/
├─ test-data/
├─ allure-results/
├─ allure-report/
├─ playwright.config.js
└─ package.json
```

Module Breakdown

Module ID	Module Name	Estimated Test Cases
M1	Homepage & Navigation	12
M2	Authentication & Registration	18
M3	Product Search Functionality	14
M4	Product Listing & Filters	15
M5	Product Details	12
M6	Cart Management	16
M7	Checkout Flow	18
M8	API & Network Validation	15
M9	Responsive & Cross-Browser UI	10
M10	Forms & Validation Testing	10

Priority Distribution Table

Module ID	Module Name	Critical	High	Medium	Low
M1	Homepage & Navigation	2	7	3	0
M2	Authentication & Registration	6	8	3	1
M3	Product Search Functionality	2	8	3	1
M4	Product Listing & Filters	2	9	4	0
M5	Product Details	3	6	3	0
M6	Cart Management	6	7	3	0
M7	Checkout Flow	8	7	3	0

M8	API & Network Validation	7	6	2	0
M9	Responsive & Cross-Browser UI	2	5	3	0
M10	Forms & Validation Testing	2	5	3	0

Priority Test Case Elaboration

Priority	Test Case ID	Title	Objective	Expected Result	Reason
Critical	TC_AU_001	Successful Login	Validate login	User logs in successfully	Core business workflow
Critical	TC_CM_005	Cart Subtotal	Validate subtotal math	Correct totals shown	Revenue impact
Critical	TC_CF_013	Payment Validation	Validate payment form	Accepts valid inputs	Purchase flow
High	TC_SR_001	Product Search	Validate search	Relevant products shown	Customer usability
High	TC_PL_003	Category Filters	Validate filtering	Filtered products displayed	Browsing experience
Medium	TC_PD_006	Image Gallery	Validate gallery	Images update correctly	UX enhancement
Low	TC_HN_010	Meta Tags	Validate metadata	SEO metadata present	Informational validation

Playwright Features Implemented

- Page Object Model (POM)
- Parallel Execution
- Cross-Browser Testing
- API Testing
- Allure Reporting
- Trace Viewer
- Screenshots & Videos on Failure
- GitHub Actions CI/CD
- Responsive Testing

- Data-Driven Testing

Allure Reporting

Allure Reports will provide execution summaries, screenshots on failure, trace analysis, flaky test detection, and historical execution trends.

EXPECTED PROJECT OUTCOME

The framework will demonstrate:

- enterprise-grade automation architecture
- scalable Playwright framework design
- modern UI automation
- API + UI hybrid testing
- Allure reporting
- CI/CD integration
- cross-browser compatibility
- professional test organization

Conclusion

This project demonstrates an enterprise-grade Playwright automation framework capable of handling UI, API, responsive, and cross-browser testing with scalable architecture and reporting integration.